

A Spectroscopic Analysis of Little Red Dots

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Abstract

The origin of Little Red Dots (LRDs) remains uncertain because their compact sizes and red optical colors can be explained by either dusty star formation or obscured active galactic nuclei (AGN). This project tests the AGN interpretation using James Webb Space Telescope (JWST) data from the JWST Advanced Deep Extragalactic Survey (JADES) Data Release 3. LRD candidates were selected using redshift, color, and compactness criteria from the NIRCам photometric catalog, resulting in 24 candidates. Of these, 22 have available JWST/NIRSpec prism and/or grating spectra and were included in the spectroscopic analysis. Emission-line fitting was performed with `PyQSOFit`, focusing on lines such as $H\beta$, $[O\ III]\ \lambda 5007$, $[Ne\ III]\ \lambda 3869$, and $[Ne\ V]\ \lambda 3426$. Because $[Ne\ V]$ is a high-ionization line associated with AGN activity, the expected $[Ne\ V]$ flux was estimated from the measured $[O\ III]$ flux and compared to the local root-mean-square (RMS) noise near the $[Ne\ V]$ wavelength. None of the 22 sources reach a 3σ $[Ne\ V]$ detection threshold. Therefore, all $[Ne\ V]$ measurements are treated as non-detections and reported as upper limits. In 15 of the 22 sources, the expected $[Ne\ V]$ flux falls within the local 1σ noise level, making these cases especially inconclusive. The remaining 7 sources show slightly more signal, with expected $[Ne\ V]$ fluxes above the local 1σ level, but they still do not reach the 3σ threshold. These results do not confirm $[Ne\ V]$ emission in the sample, but they place upper-limit constraints on possible AGN activity in LRDs. These upper limits are crucial to clarify what the current spectra can and cannot rule out about obscured AGN activity in LRDs.

1 Introduction

One open question in high-redshift astronomy is how black holes grew so quickly in the early universe. The *James Webb Space Telescope* (JWST) has revealed a population of compact red sources, known as Little Red Dots (LRDs), at redshifts $z \approx 4\text{--}7$. These objects are interesting because they appear only $\sim 600\text{ Myr--}1.5\text{ Gyr}$ after the Big Bang, when the first galaxies and black holes were still forming. Their physical nature is still debated because their photometry can be explained in more than one way.

A key feature of LRDs is their V-shaped spectral energy distribution (SED), with a blue ultraviolet (UV) slope and a red optical slope. This shape could come from dusty star formation, where dust changes the observed galaxy colors (Casey et al., 2024). It could also come from obscured active galactic nuclei (AGN), where accretion onto a black hole contributes to the observed emission (Inayoshi, 2025). Observational studies have also suggested a possible connection between LRDs and AGN activity, including evidence from spectroscopy and X-ray analysis that some brighter LRDs may host moderately obscured AGN (Kocevski et al., 2024). At the same time, heavily obscured AGN can be missed even in multiwavelength surveys (Peca et al., 2025), which makes it difficult to separate star formation from accretion in faint, high-redshift sources.

The primary goal of this project is to determine whether LRDs host obscured AGN by searching for high-ionization spectral signatures, particularly the [Ne V] $\lambda 3426$ emission line, which is extremely difficult to produce through stellar processes alone since it requires very high-energy photons. As a result, it serves as a clean tracer of AGN activity.

It is important to note that not all AGN necessarily show detectable [Ne V] emission in the available spectra. While [Ne V] is a clean AGN tracer, its absence does not automatically mean that a source is not an AGN. The line may be intrinsically weak, obscured, or hidden by the noise level of the observation. For this reason, this project does not only ask whether [Ne V] is directly detected, but whether [Ne V] should have been detectable given the quality of the data. This distinction is important because it separates true non-detections from inconclusive cases where the spectra are not sensitive enough.

Determining whether LRDs host obscured AGN is critical for understanding how supermassive black holes formed and grew so rapidly within the first billion years after the Big Bang. Because LRDs may represent a short-lived evolutionary phase in early galaxy or AGN evolution, they could reveal a previously hidden population of actively accreting black holes. Identifying AGN activity in LRDs is therefore essential for understanding the connection between black hole growth, dust obscuration, and galaxy evolution at high redshift.

2 Data

This study uses archival data from the JWST Advanced Deep Extragalactic Survey (JADES) Data Release 3 in the GOODS-North and GOODS-South fields. The analysis combines imaging data and photometric catalogs from the Near Infrared Camera (NIRCam) with spectroscopic observations from the Near Infrared Spectrograph (NIRSpec). The NIRSpec spectra include both low-resolution prism observations and higher-

resolution grating modes (F070LP/G140M, F170LP/G235M, and F290LP/G395M), allowing the analysis to use both broad wavelength coverage and more detailed emission-line measurements.

These data products were accessed through the JADES catalogs and the Mikulski Archive for Space Telescopes (MAST). The spectra cover key emission lines needed for this project, including $H\beta$, $[O\ III]\ \lambda 5007$, $[Ne\ III]\ \lambda 3869$, and $[Ne\ V]\ \lambda 3426$. This combination of imaging, photometry, and spectroscopy makes JADES DR3 useful for selecting LRD candidates and testing whether their spectra show evidence for high-ionization emission associated with AGN activity.

A three-step selection process was used to identify LRD candidates: redshift selection, color selection, and compactness selection.

Selected sources had $z > 4$ from spectroscopic catalogs. This redshift cut corresponds to the epoch of early galaxy and black hole growth where LRDs are predominantly observed in JWST surveys. This also ensures that the observed photometry probes rest-frame UV and optical wavelengths relevant for identifying the characteristic LRD spectral energy distributions.

Following [Kocevski et al. \(2024\)](#), UV and optical slopes (β) were computed using:

$$\beta = 0.4 \cdot \frac{m_1 - m_2}{\log_{10}(\lambda_2/\lambda_1)} - 2, \quad (1)$$

where m_1 and m_2 are the AB magnitudes measured in two filters and λ_1 and λ_2 are the corresponding filter wavelengths. AB magnitudes are a logarithmic way of measuring brightness based on flux density, so smaller AB magnitudes correspond to brighter sources. Photometric fluxes from the JADES catalog were given in nanojanskys (nJy) and were converted to AB magnitudes using:

$$m_{AB} = -2.5 \log_{10}(f_{nJy}) + 31.4, \quad (2)$$

where f_{nJy} is the measured flux density in nanojanskys.

Candidates were selected using $\beta_{UV} < -0.37$, $\beta_{UV} > -2.8$, and $\beta_{opt} > 0$. This isolates the characteristic V-shaped spectral energy distributions associated with LRDs. The cut $\beta_{UV} > -2.8$ is used to remove possible brown dwarf contamination, while $\beta_{opt} > 0$ selects sources with a red optical slope. In practice, this selection identifies sources with a blue UV slope and a rising red optical slope. Figure 1 shows the color-selection step used to identify sources with the characteristic LRD continuum shape.

To further isolate compact, point-like sources, a morphological selection was applied based on the F444W filter, which probes rest-frame optical emission at these redshifts. A stellar locus was constructed using F444W magnitude as a function of half-light radius. The F444W filter was chosen because, at $z \geq 4$, it probes rest-frame optical wavelengths, providing a more reliable tracer of galaxy structure that is less affected by dust attenuation and offers higher signal-to-noise for faint sources.

Following the compactness criteria used by [Kocevski et al. \(2024\)](#), sources were classified as point-like if they lay within $1.5\times$ the stellar locus and satisfied a signal-to-noise ratio cut of $SNR > 12$. This step ensures that selected objects are consistent with unresolved sources, as expected for LRDs. Figure 2 shows the compactness selection used to keep point-like sources and remove extended objects.

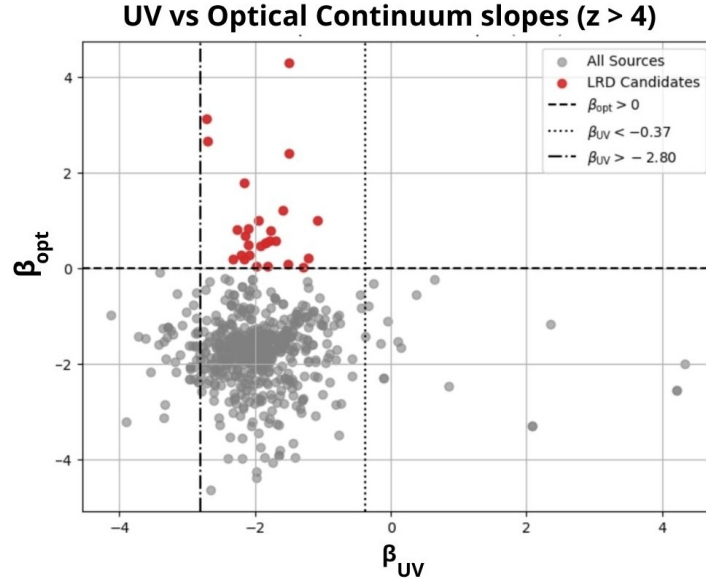


Figure 1: Color-based selection of Little Red Dots (LRDs) using rest-frame UV and optical continuum slopes from the JADES survey. Gray points show the full source sample, while red points show the selected LRD candidates. The dashed and dotted lines indicate the color cuts in β_{UV} and β_{opt} .

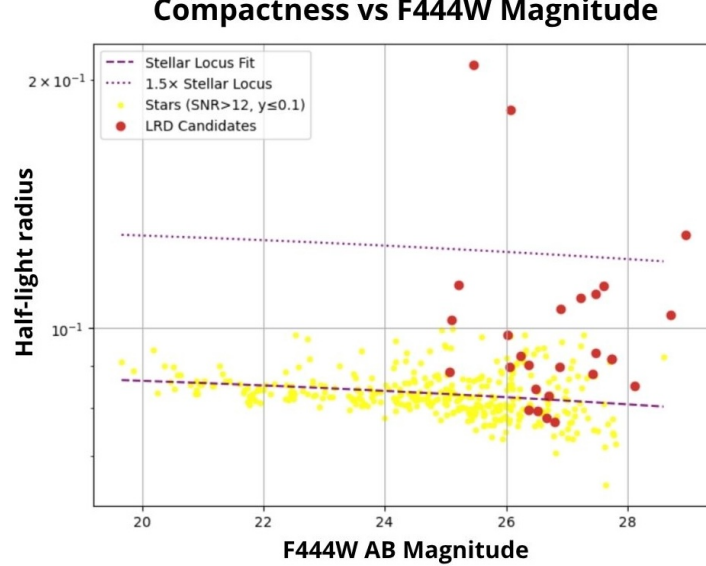


Figure 2: Compactness selection using F444W AB magnitude and half-light radius. The dashed purple line shows the stellar locus fit, while the dotted purple line marks the $1.5\times$ stellar-locus threshold used for the cut. Yellow points indicate stars used to define the locus, and red points indicate sources that satisfy the LRD color selection before the compactness cut. Red sources below the dotted line are classified as compact, while sources above the dotted line are more extended and are not included in the final compact LRD sample.

This selection resulted in a total of 24 LRD candidates. Two of these candidates do not have corresponding spectral files in the JADES DR3 products, so the spectroscopic analysis was performed on the remaining 22 sources with available spectra.

3 Methods and Analysis

Python-based tools were used throughout the analysis. `Astropy` was used to handle astronomical data products, including reading spectral files and working with units and tables. `NumPy` was used for numerical calculations, including flux ratios, RMS noise estimates, and logarithmic line ratios. `Matplotlib` was used to create the selection plots and emission-line diagnostic diagram. These tools allowed the photometric and spectroscopic measurements to be processed before the final line-ratio analysis.

Spectra for the 22 sources were fitted using `PyQSOFit`, a spectral-fitting package designed to model active galactic nucleus (AGN) spectra (Guo et al., 2018). The fitting targeted the main emission lines relevant to this project, including $H\alpha$, $H\beta$, $[O\ II]\ \lambda 3727$, $[O\ III]\ \lambda 5007$, $[Ne\ III]\ \lambda 3869$, $[Ne\ V]\ \lambda 3426$, and $He\ II\ \lambda 4686$.

`PyQSOFit` models the continuum and emission-line components of each spectrum and returns best-fit line fluxes with associated fitting uncertainties when available. The fitting procedure can use Markov Chain Monte Carlo (MCMC) sampling to estimate the range of possible model parameters and the corresponding uncertainties on line fluxes. In this project, the fitted $[O\ III]\ \lambda 5007$, $[Ne\ III]\ \lambda 3869$, $[Ne\ V]\ \lambda 3426$, and $H\beta$ fluxes were used for the main analysis. The $[O\ III]$ measurements were used to estimate the expected $[Ne\ V]$ flux, while $[Ne\ III]$, $[Ne\ V]$, $[O\ III]$, and $H\beta$ were used to construct the emission-line diagnostic diagram.

For each source, emission-line fluxes were extracted for $[O\ III]$, $[Ne\ III]$, and, where present in the fit, $[Ne\ V]$. Since a non-detection of $[Ne\ V]$ does not rule out the presence of an AGN, the analysis focuses on detectability and upper limits rather than treating every non-detection as evidence against AGN activity.

To test whether $[Ne\ V]$ should have been detectable, the measured $[O\ III]\ \lambda 5007$ flux was used to estimate the expected $[Ne\ V]$ flux using the assumed relation:

$$F_{[NeV]} \approx 0.05 \times F_{[OIII]}, \quad (3)$$

where $F_{[NeV]}$ is the expected $[Ne\ V]$ line flux and $F_{[OIII]}$ is the measured $[O\ III]\ \lambda 5007$ line flux. This expected $[Ne\ V]$ flux was then compared to the local root-mean-square (RMS) noise near the expected $[Ne\ V]$ wavelength.

The local RMS noise near $3426\ \text{\AA}$ was computed using:

$$\text{RMS} = \sqrt{\langle f^2 \rangle}, \quad (4)$$

where f represents the continuum-subtracted flux density near the $[Ne\ V]$ line. The RMS gives a local estimate of the noise level near the wavelength where $[Ne\ V]$ would be expected.

The expected $[Ne\ V]$ flux was compared to the local 1σ , 2σ , and 3σ noise levels. A signal above 1σ is not strong enough to claim a detection, because random noise fluctuations can commonly produce this level of significance.

To improve constraints on emission-line measurements, spectra were examined across different NIRSPEC observing modes, including low-resolution prism data and higher-resolution grating modes (G140M, G235M, G395M). The prism provides broad wavelength coverage, while the gratings offer improved spectral resolution over narrower wavelength ranges, allowing for more precise measurements in specific spectral regions. To further investigate the nature of the sources, an emission-line diagnostic diagram was constructed using ratios of key spectral lines. In particular, $\log([\text{Ne V}]/[\text{Ne III}])$ and $\log([\text{O III}]/\text{H}\beta)$ were computed using integrated fluxes with propagated uncertainties when available. The $[\text{Ne III}]$, $[\text{O III}]$, and $\text{H}\beta$ fluxes provide useful measurements of the ionization state of the gas.

4 Results

The $[\text{Ne V}]$ analysis did not yield statistically significant detections. None of the 22 sources reach a 3σ detection threshold relative to the local 1σ RMS noise near the expected $[\text{Ne V}]$ wavelength. Therefore, all $[\text{Ne V}]$ measurements are treated as non-detections and reported as upper limits.

The sources can still be separated by how close the expected $[\text{Ne V}]$ flux is to the local noise level. In 15 of the 22 sources, the expected $[\text{Ne V}]$ flux falls within the local 1σ RMS noise level. These sources are especially inconclusive because the spectra are not sensitive enough to determine whether $[\text{Ne V}]$ is absent or simply hidden by noise. In the remaining 7 sources, the expected $[\text{Ne V}]$ flux is above the local 1σ RMS level, meaning there is slightly more signal relative to the noise. However, because these sources do not reach 3σ , they are not considered detections. A 1σ excess is not enough to claim a detection because random noise fluctuations can produce this level of significance with a relatively high probability.

The main result is therefore not a confirmed detection of $[\text{Ne V}]$, but a set of upper limits on $[\text{Ne V}]$ emission in the LRD sample. These upper limits are important because they show how strong emission could be without being significantly detected in the current spectra. This matters because a non-detection does not automatically rule out an AGN; it may mean that the spectra are not sensitive enough to detect the expected $[\text{Ne V}]$ emission.

Figure 3 shows the resulting emission-line ratio diagram. The $[\text{Ne III}]$, $[\text{O III}]$, and $\text{H}\beta$ measurements provide useful information about the ionization state of the gas. However, because $[\text{Ne V}]$ is not detected at the 3σ level, the $[\text{Ne V}]/[\text{Ne III}]$ ratios are upper limits. This means that the plotted x-values should not be interpreted as exact measurements of the true $[\text{Ne V}]/[\text{Ne III}]$ ratios. The true $[\text{Ne V}]$ flux could be lower, which would move sources leftward in the diagram. As a result, sources that appear near AGN-like or composite regions could also be consistent with composite or star-forming regions depending on the true $[\text{Ne V}]$ flux.

The BPT-like diagram does not confirm AGN activity, but it remains useful. It shows which sources are still consistent with AGN-like or composite ionization within the current upper-limit constraints. By combining sample selection, $[\text{Ne V}]$ upper limits, and emission-line ratios, this project places constraints on possible AGN activity in LRDs even without a statistically significant $[\text{Ne V}]$ detection.

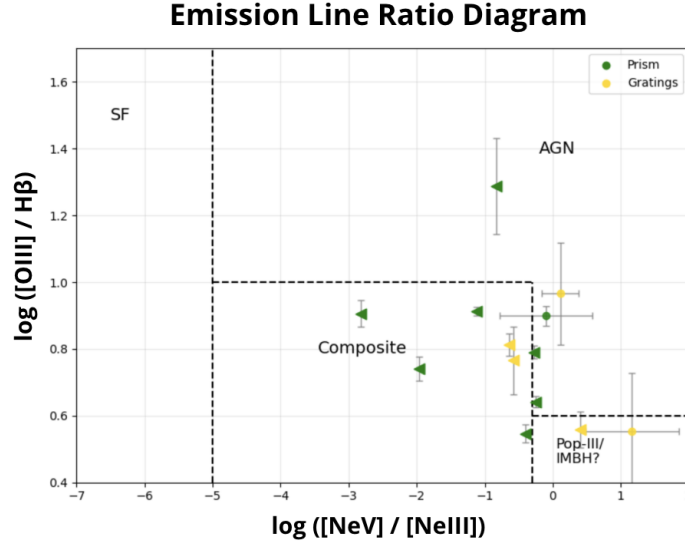


Figure 3: Emission-line ratio diagram using $\log([Ne V]/[Ne III])$ and $\log([O III]/H\beta)$. Green points show prism spectra, while yellow points show grating spectra. The $[Ne III]$, $[O III]$, and $H\beta$ lines were measured when available. When a source is plotted as a point with error bars, the $[Ne V]$ -based ratio should be interpreted as a constraint rather than a confirmed $[Ne V]$ detection. Triangular markers indicate sources plotted as upper limits.

5 Conclusions

This analysis tested whether Little Red Dots host obscured AGN by searching for the high-ionization $[Ne V] \lambda 3426$ emission line in JWST/NIRSpec spectra from JADES DR3. A three-step selection process was used to identify LRD candidates using redshift, color, and compactness criteria. This selection resulted in 24 LRD candidates, of which 22 have available spectra and were included in the spectroscopic analysis.

Spectra for the 22 sources were fitted using `PyQSOFit`, targeting key emission lines including $H\alpha$, $H\beta$, $[O II] \lambda 3727$, $[O III] \lambda 5007$, $[Ne III] \lambda 3869$, $[Ne V] \lambda 3426$, and $He II \lambda 4686$. To test for $[Ne V]$ detectability, the expected $[Ne V]$ flux was estimated from the measured $[O III]$ flux and compared to the local RMS noise near $3426 \text{ \AA}$.

Overall, 15 sources are consistent with the local noise, while 7 show expected $[Ne V]$ levels above 1σ but below the 3σ detection threshold.

These non-detections are still scientifically important. Upper limits provide constraints on how strong $[Ne V]$ emission could be in each source without being detected. When combined with measured $[Ne III]$, $[O III]$, and $H\beta$ fluxes, these limits help test whether the line ratios are consistent with AGN-like ionization or whether the sources could also fall in composite or star-forming regions.

Therefore, this project does not confirm $[Ne V]$ emission or prove that the LRDs host AGN. Instead, it places upper-limit constraints on possible $[Ne V]$ emission and shows that the current spectra are limited by signal-to-noise. These constraints are important because they prevent over-interpreting non-detections and clarify what the available data can and cannot say about AGN activity in LRDs. Even without robust detections, measuring these limits is a necessary step toward understanding whether

compact high-redshift LRDs are connected to early black hole growth.

Future work could build on these upper limits. One next step would be stacking spectra to increase the signal-to-noise ratio and test whether weak [Ne V] emission is present below the detection limit of individual spectra. Another step would be expanding the sample beyond JADES, for example by including LRD candidates from the Cosmic Evolution Early Release Science Survey (CEERS). A larger sample would help test whether the trends seen here are specific to these 22 sources or more representative of the broader LRD population.

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